



Research paper

Influence of the cell size and wall thickness on the compressive behaviour of fused filament fabricated PLA gyroid structures

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ABSTRACT

The development of Additive Manufacturing (AM) has greatly facilitated the fabrication of cellular and lattice materials. Gyroid-based lattice structures, known for their distinctive properties such as interconnected porosity, high surface-to-volume ratio, and remarkable structural stiffness combined with specific energy absorption, have been extensively explored. Many studies examining the impact of design parameters on the mechanical properties of Gyroid lattice materials have utilized metal AM techniques. This research aims to evaluate the influence of two design parameters on the compressive properties of Gyroid structures obtained by fused filament fabrication (FFF). A full factorial analysis was employed to assess the effects of cell size and wall thickness on the compressive properties of polylactic acid (PLA) Gyroid lattices. Cell sizes were varied between 4 mm, 5 mm, and 10 mm, while wall thickness ranged from 0.4 mm, 0.6 mm, to 0.8 mm. After 3D printing, the print quality was assessed, samples were weighted and then subjected to compression testing. During compression, the lattices with 10 mm cells exhibited successive layer collapse, whereas the lattice with 4 mm and 5 mm cells displayed plastic deformation, marked by a plateau in the stress-strain curve. These behaviours were mostly independent of wall thickness, except for the 5 mm cell lattice with 0.4 mm wall thickness. The elastic modulus, yield stress and absorbed energy per volume aligned with the apparent density of the lattices, ranging between 1% and 12% of the bulk 3D printed material for both the stiffness and yield stress, and between 1% and 22% for the energy absorbed. Analysis of the fitted means indicated that doubling the cell size had a more significant impact on the measured properties than doubling the wall thickness, while doubling both the cell size and wall thickness exerted a more pronounced influence on the yield stress and strain. Notably, under the conditions of this study, the 3D printed PLA Gyroids behaved similarly to closed cell foams, despite their interconnected channels. Their compressive mechanical properties comparable to those of rigid polyurethane foams with closed cells.

1. Introduction

Cellular structures differ from other types of hybrid materials (such as composites, sandwich and segmented structures) in that their final properties are not solely determined by material combinations. Instead, the primary factor of influence is the efficient utilization of space (Ashby, 2011). Cellular solids can be classified based on the structural organization level they possess. These include those with random closed and open cells, represented by foams, and those with periodic architectures, known as lattice structures (Evans et al., 2001). It's also possible to classify cellular solids based on their predominant mechanical behaviour, categorizing them into bending- or stretch-dominated structures. Typically, in foam structures, the cell edges tend to bend under loading, whereas in lattice structures, the trusses or struts

primarily stretch under loading (Ashby, 2006).

In contrast to foams, where cell sizes, distribution, and properties are determined by physical-chemical phenomena used to trap gases or fluids in the solid, lattice structures can be entirely designed. High level control over the cell topology and lattice density allows for the induction of unique properties that diverge from those of their constituent materials and from those commonly found in nature. In such cases, the term 'metamaterial' is applied. The properties of metamaterials include, among others, auxetic behaviours and high specific strength (Makatura et al., 2023).

Metamaterials based on triply periodic minimal surfaces (TPMS) are a specific type of *shellular* structure whose unit cells are formed by zero mean curvature surfaces that repeat themselves in three dimensions, dividing a given volume into two or more intertwining but disjointed

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continuous channels (Makatura et al., 2023), (Wallat et al., 2022). From the various TPMS described in literature, the Schoen's Gyroid is one of the most common surfaces explored in 3D printed metamaterials (Chouhan and Bala Murali, 2023). Some of the main features of interest in Gyroid lattices include their interconnected channels that favour mass transport processes, essential for cellular growth in tissue engineering scaffolds (Chouhan and Bala Murali, 2023), (Zadpoor, 2019), (Zhu et al., 2022), (Kelly et al., 2021); the high surface-to-volume ratio and complex channel geometry that result in higher heat transfer rates, which is much sought after for heat exchanging applications (Wallat et al., 2022), (Chouhan and Bala Murali, 2023), (Bonner-Hutton et al., 2023); and their high structural stiffness and low maximum von Mises stress, of interest for high specific energy absorption applications (Maskery et al., 2017).

A Gyroid can be obtained from trigonometric approximations, as expressed in Eq. (1), in terms of the coordinates (x, y, and z), the size of the unit cell (a), and the isovalue (t) parameter that determines the cross-section of the surfaces (Tino et al., 2019). The size of the unit cell (a) can be calculated by dividing the overall size of the lattice structure by the number of repeating unit cells.

$$\sin\left(\frac{2\pi}{a}x\right)\cos\left(\frac{2\pi}{a}y\right) + \sin\left(\frac{2\pi}{a}y\right)\cos\left(\frac{2\pi}{a}z\right) + \cos\left(\frac{2\pi}{a}x\right)\sin\left(\frac{2\pi}{a}z\right) = t \quad (1)$$

While the isovalue (t) plays a crucial role in the resulting shape of the Gyroid, it is typically set to zero to generate the outermost surface with zero mean curvature. Another essential design parameter of solid Gyroid structures is wall thickness, which is often introduced by offsetting the original surface in both directions (Tino et al., 2019). Besides the geometry of the structure, the mechanical properties of the Gyroid lattices are strongly influenced by their apparent density (ρ^*), i.e., the ratio between the mass of solid material and total volume occupied by the lattice, which depends on the chosen design parameters (Wallat et al., 2022).

Since the 2000s, with the rapid development of additive manufacturing (AM) technologies, the production of complex lattice metamaterials has been highly facilitated, when compared with their production using conventional manufacturing methods (Maharjan et al., 2018), (Agnelli et al., 2023). The review by Nazir et al. (2019) provides a comprehensive analysis of various lattices produced with AM, with insights on the design, optimization, and applicability of these structures. Benedetti et al. (2021) summarise types of lattice structure-based literature, with a special emphasis on the identification of AM-related defects and their fatigue and fracture behaviour.

Studies on the influence of lattice design parameters on the mechanical behaviour of 3D-printed Gyroid structures were conducted employing selective laser melting (SLM) processes (Maskery et al., 2017), (Yang et al., 2019), (Mahmoud et al., 2021). SLM is a powder bed fusion (PBF) AM technology, in which material particles are selectively fused by a laser source (ISO/ASTM 52900, 2015), enabling the reproduction of intricate and fine-detail geometries (Kelly et al., 2021), (Yang et al., 2019), (Mahmoud et al., 2021). In their work, Maskery et al. (2017) investigated the failure modes of Al-Si10-Mg Gyroid lattices. They varied the size of the unit cell while keeping a constant volume fraction. The lattices with the smallest cells exhibited failure through diagonal shear, while the lattices with larger cells experienced failure via crack propagation parallel to the applied load. The lattices with intermediate cell sizes failed due to the successive collapse of the Gyroid planes, perpendicular to the applied load. In a comprehensive numerical investigation, Yang et al. (2019) explored the effect of lattice parameters, including cell size, wall thickness, number of cells, and isovalue, on the compressive properties of Ti-6Al-4V Gyroid structures produced by SLM. Results indicated that higher compressive strengths could be achieved by increasing the wall thickness, which was shown to be proportional or even more effective than reducing the cell size. It was also

found that by increasing the number of cells, the extent of elastic behaviour was reduced to shorter strains even though the overall structural strength increased. In addition, increasing the isovalue promoted a transition from high stiffness to mostly plastic behaviour accompanied by a sensible reduction of the structural strength. Mahmoud et al. (2021) analysed how SLM defects influenced the fatigue properties of Ti-6Al-4V Gyroid lattices within cylindrical samples. Three design variations for the same value of relative density, including samples with thinner and thicker walls, as well as samples with radially graded wall thickness, were assessed. The study revealed that the thicker walls led to a higher incidence of internal defects due to heat accumulation, subsequently reducing the number of cycles required for crack initiation.

Fused Filament Fabrication (FFF) is considered one of the most accessible, flexible, and cost-effective AM techniques for processing thermoplastics (Bouzaglou et al., 2023). In FFF an extrusion head system is attached to a carriage, where the material is heated to a semi-molten state and selectively deposited onto a build platform, or a previously deposited layer of material (Costa Oliveira et al., 2023). Extensive literature exists on the behaviour of cellular structures obtained by FFF. Oliveira et al. (2022) tuned FFF process parameters to produce cubic and sandwich structures with spherical TPMS unit cells, identifying production defects associated with filament stringing and retraction. When it comes specifically to Gyroid-type TPMS, the manufacturability and performance of these structures as radiation dosimetry phantoms was assessed by Tino et al. (2019). Different lattices were designed varying the number of unit cells and wall thicknesses, and subsequently printed using acrylonitrile butadiene styrene (ABS). The defects, however, had little or no effect on the measured radiodensities. The radiodensity was found to increase with the wall thickness and number of unit cells (Tino et al., 2019). Zhu et al. (2022) analysed the compressive modulus of polycaprolactone (PCL) Gyroid scaffolds, comparing designs with and without additional hole matrices. The lattice design parameters were kept constant and only the size of the supplementary holes varied to enhance gas diffusion and mass transport capabilities. It was found that the elastic modulus of the scaffolds with additional hole matrices more closely approximated to that of the breast tissue were similar to the standard Gyroid scaffolds. In the study conducted by Holmes et al. (2022), Gyroid lattices with 10, 15, and 20 mm cells and wall thickness of 0.6 and 1.2 mm were 3D printed using thermoplastic elastomers. The objective was to investigate whether these flexible structures could be used as tuneable substitutes for polyurethane (PU) foams. Besides establishing power law relationships between compressive properties and apparent density, the study evaluated the performance of the flexible lattices under cyclic loading. The results demonstrated comparability with the behaviour of PU foams commonly used in rehabilitation cushions.

The effects of the design parameters on the mechanical properties of FFF printed Gyroid structures was first investigated by Maharjan et al. (2018). They characterized the compression behaviour of polymer-based Gyroid structures with 6, 8, 10, and 12 mm unit cells, designed with 15%, 20%, and 25 % volume fraction. The samples were printed using polylactic acid (PLA). In general, the compressive strength tended to decrease with larger unit cells at a same relative volume, whereas for the same cell sizes the compressive strength increased with the relative volume. As observed by the authors, even so Gyroid structures with smaller cells and higher relative volume result in better mechanical properties, the time required to fabricate denser lattices with finer unit cells is always longer (Maharjan et al., 2018). In a similar work, Machonachie et al. (Maconachie et al., 2020) investigated the effect of the number of cells and wall thickness on manufacturability and mechanical properties of ABS Gyroid structures. Lattices were designed with 2, 3, 6, 7, and 8 cells, with wall thicknesses ranging from 0.5 mm to 3 mm. The main findings pointed out to improved mechanical performance with the number of cells and wall thickness, the latter having a more significant effect on the results. Even though not all design

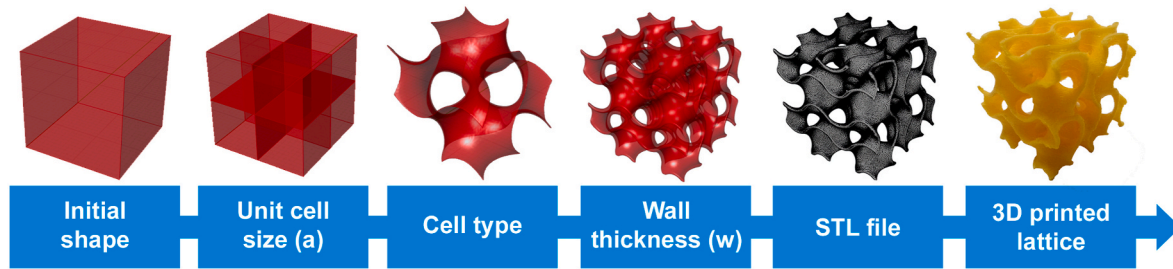


Fig. 1. Design steps used to create the Gyroid lattices.

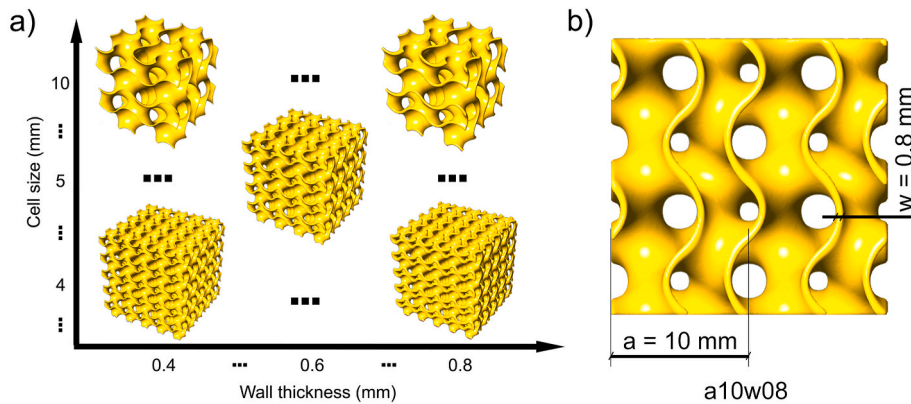


Fig. 2. (a) Effect of the design parameters in the resulting Gyroid lattices, (b) general labelling scheme.

combinations were explored, the Gibson-Ashby equations generated from the regression analysis demonstrated strong correlations in predicting relative stiffness and strength in function of the relative density.

In this paper, the authors conduct a full factorial three-level analysis to explore the influence of cell size (a) and wall thickness (w) on the compression behaviour of 3D printed PLA Gyroid structures. While decreasing the cell size or increasing wall thickness is expected to proportionally increase the apparent density of the lattice and, consequently, improve the mechanical properties, it is crucial to determine which factor has the greatest influence on the results. Different from the research conducted by Maharjan et al. (2018), where the value for wall thickness is determined by the combination of cell size and relative density, designing the lattices with prescribed cell size and wall thickness allows having two or more distinct structures with the same relative density. Additionally, it would be interesting to know if the mechanical behaviour of these structures can be predicted by Gibson-Ashby equations (Ashby, 2006). Even though a similar study was made by Machonachie et al. (Maconachie et al., 2020), the material used by authors, ABS, is significantly tougher than PLA. Therefore, our study aims to provide insights into understanding the primary contributors to the elastic modulus, yield stress and strain, and absorbed energy of fused filament fabricated PLA Gyroids, enhancing the overall applicability of these structures.

2. Materials and methods

Gyroid lattices were generated using the Crystallon® plugin for Rhino® 7 in the Grasshopper® environment. This plugin enables the creation of cellular structures by voxelizing a specified volume and assigning the chosen unit cell type to each voxel. The voxel dimensions are user-defined, and a range of unit cell options are available including strut-based and shell-based cells. For this study, a 20 mm cube was voxelized according to different values of cell size (a). Lattices were formed by filling each voxel with $t = 0$ Gyroid surfaces. Subsequent to lattice creation, the surfaces were thickened according to the desired

Table 1
3D printing parameters used to all topologies.

Process Parameter	Value
Nozzle temperature [°C]	205
Bed temperature [°C]	60
Layer height [mm]	0.1
Extrusion width [mm]	0.4
Infill density [%]	100
Printing speed [mm·s ⁻¹]	50
Retraction speed [mm·s ⁻¹]	40
Retraction distance [mm]	1.5

wall thickness (w). The resulting structures were then exported as stereolithography (STL) files to be 3D printed. The design workflow is schematically represented in Fig. 1.

Given the flexibility of changing the lattice structure through parametrized design, the volume fraction of virtual Gyroid structures was initially evaluated across a range of lattice designs. Cells sizes (a) varied from 3 mm to 10 mm, while wall thicknesses (w) ranged from 0.4 mm to 0.8 mm, resulting in 40 lattice designs. Fig. 2a illustrates how these two parameters influenced the resulting lattice designs. For the 3D printed samples, the criteria were established to ensure only complete Gyroids were present in all three directions of the 20 mm cube. Therefore, structures with cells of 10 mm, 5 mm, and 4 mm, were selected. With respect to the wall thickness, the aim was to select values corresponding to the extrusion of two, one and half, and one strand produced by a 0.4 mm FFF nozzle. Consequently, designs with wall thickness of 0.8 mm, 0.6 mm, and 0.4 mm wall thickness were chosen for 3D printing. The general labelling scheme for the 3D printed structures is depicted in Fig. 2b, where, for instance, the label “a10w08” denotes a structure with 10 mm Gyroids cells thickened by 0.8 mm.

Five specimens were 3D printed for each of the nine lattices design selected, using a yellow PLA filament with 1.75 mm diameter procured from Filkemp® and a Raise3D® Pro 2 3D printer. Uniform 3D printing parameters were applied to all samples, as outlined in Table 1.

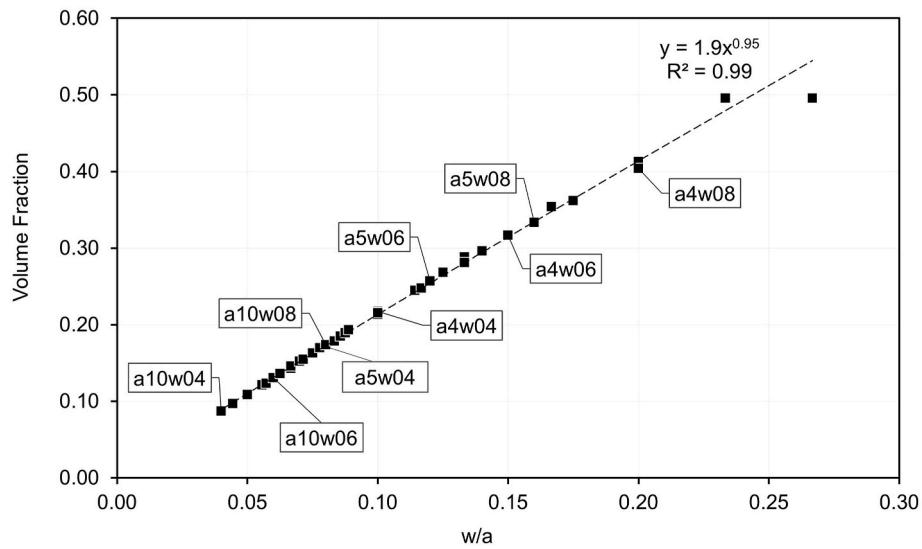


Fig. 3. Volume fraction in function of the w/a ratio for the different lattice designs. The labels indicate the cell size and wall thickness combinations selected to be 3D printed.

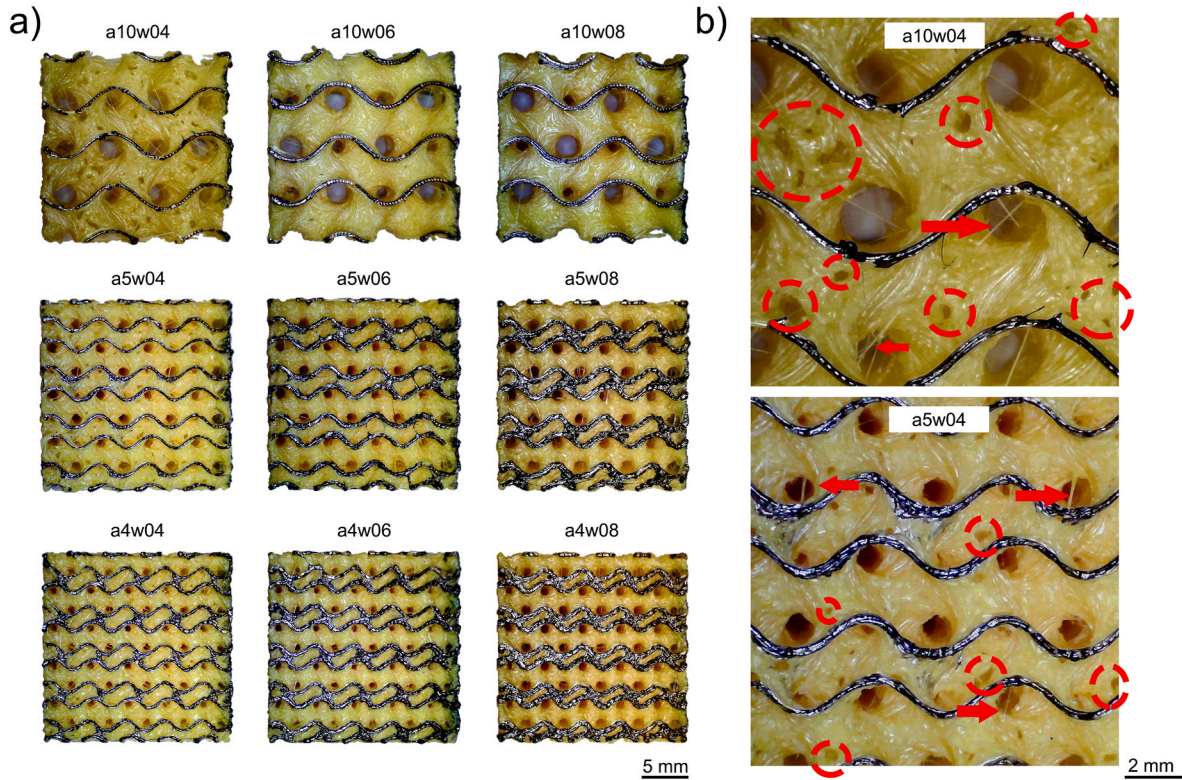


Fig. 4. (a) Representative samples of the 3D printed Gyroid lattices, (b) detailed view of two structures showing the main 3D printing defects: wall holes (circles) and strings (arrows).

The apparent density (ρ^*) of the 3D printed lattices was calculated by measuring the mass of the samples and dividing by the volume of the cube. An analytical Kern Pcb 1000-2 scale was used to measure the mass of five samples per topology, as recommended by the ASTM D1622-20 standard. The edges forming the top and lateral faces of the lattice structures were painted with a black permanent marker, in order to facilitate visualization. Images of the top faces were captured using a digital microscope by Velleman®, model CAMCOLMS1N, and used to identify possible 3D printing defects or irregularities.

The compression tests were conducted at ambient temperature

(21 °C) using an Instron® 5566 universal testing machine. A displacement rate of 2 mm/min was applied, consistent with the ASTM D1621-16 recommendation of 10% of the specimen's height per minute. The Gyroid lattices were compressed along the build direction until they reached a final height of 7.5 mm (62.5% strain) to avoid exceeding the maximum load allowed for the 10 kN load cell. The deformation of a lateral face of the lattices was recorded using industrial cameras (Stingray®, from Allied Vision GmbH), setting up the Vic-Snap® 2012 software to capture images at a rate of one per 8 s. Statistical analysis of the results was performed using Minitab® software.

Table 2

Measured and expected weight, weight deviation, and apparent density of the Gyroid lattices.

Lattice Design	Mass (g)		Mass Deviation (%)	ρ^* (g/cm ³)	
	Measured	Expected		Measured	Expected
a10w08	1.64	1.73	4.82	0.21	0.22
a10w06	1.26	1.29	2.52	0.16	0.16
a10w04	0.80	0.86	7.56	0.10	0.11
a5w08	3.07	3.31	7.11	0.38	0.41
a5w06	2.48	2.55	2.57	0.31	0.32
a5w04	1.60	1.72	7.09	0.20	0.22
a4w08	3.69	4.01	7.88	0.46	0.50
a4w06	3.11	3.14	1.10	0.39	0.39
a4w04	2.10	2.15	2.35	0.26	0.27

3. Results and discussion

The volume fraction for the 40 Gyroid lattices with different combinations between cell size and wall thickness created with the parametrized design is shown in Fig. 3. By plotting in terms of w/a ratio, the fitted trend exhibits a nearly a linear relationship (exponent 0.95) with an R-squared value of 0.99.

Since the volume fraction is equivalent to the relative density (ρ^*/ρ_s), a linear Gibson-Ashby relationship can be suggested. As expressed by Eq. (2), the relative density of the Gyroid lattices can be predicted as other closed-cell structures in terms of the wall thickness (w) and cell size (a), with C representing a constant. This is in agreement with the findings from Maconachie et al. (2020) for relative densities up to 0.2.

$$\frac{\rho^*}{\rho_s} = C \left(\frac{w}{a} \right) \quad (2)$$

Fig. 4a displays representative samples of the 3D printed lattices. The main printing defects are highlighted in Fig. 4b, which encompass both excessive and insufficient material deposition. Among the defects related with excess material, stringing is the most prevalent, arising due to ineffective retraction during travel movements of the extrusion head within the Gyroids' channels. Additionally, blobs or areas with excess of extruded material are visible and can be linked with deceleration in regions where the printing direction suddenly changes. Conversely, wall holes, particularly in areas where the structure is overhanging, represent the most noteworthy defects associated with insufficient material deposition. These holes result from the inability of the extruded beads, which have ellipsoidal cross-section (Bayati et al., 2024), to accurately replicate the thinner Gyroid surfaces at the specified layer height and width.

Similar defects have been identified in previous works involving Gyroid structures printed with various materials. In ABS structures produced by Maconachie et al. (2020), the presence of wall holes in unsupported surfaces, particularly in less dense designs, is prominent. In the study by Holmes et al. (2022), which utilized thermoplastic polyurethanes (TPUs), both wall holes and strings are observed, with the latter being more prevalent in TPUs compared to PLA.

The aspect of the printed samples highlights the significance of the relationship between wall thickness and cell size concerning the limitations of the FFF process. With a fixed set of printing conditions, the toolpath generated tends to more accurately reproduce the virtual shape of larger cells. However, as the cell size increases, so does the size of the unsupported surfaces. When the relative density becomes too low (very

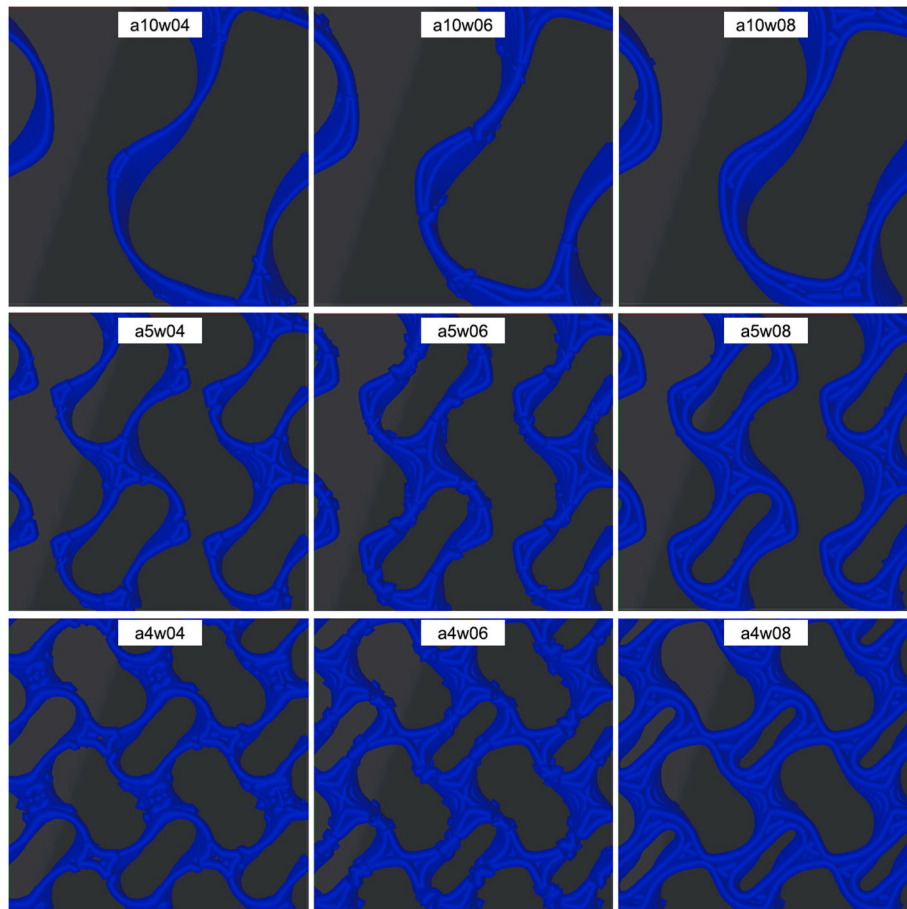


Fig. 5. Previews of the toolpaths generated by the slicing software for each Gyroid topology.

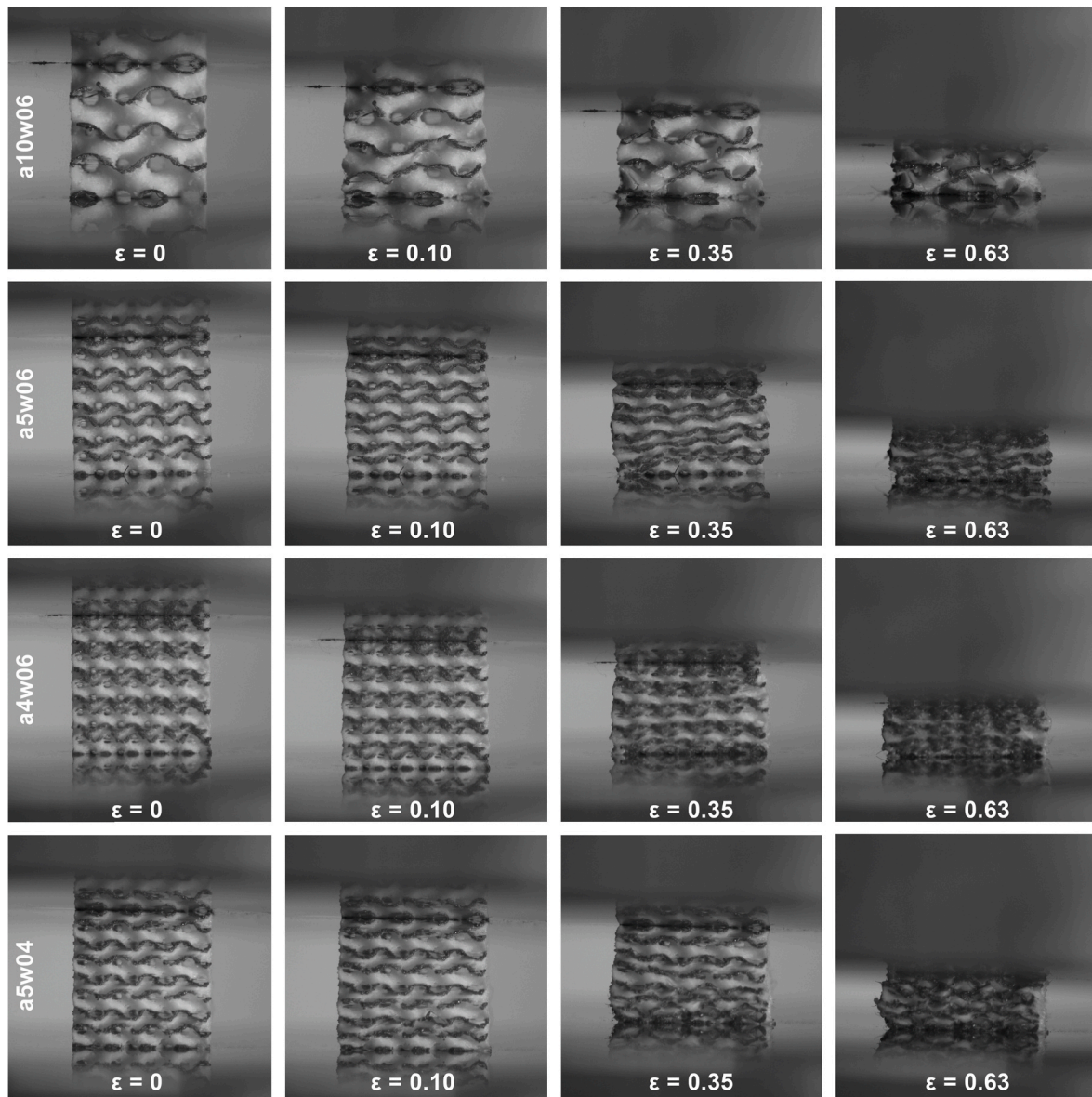


Fig. 6. Evolution of the compressive behaviour of selected Gyroid lattices.

thin walls compared to the cell size), this can lead to the formation of wall holes. Conversely, although wall holes are less likely to form with smaller cells, it becomes progressively challenging for the process to replicate the curvature of the smaller Gyroid cells, ultimately resulting in a deterioration of the lattice geometry.

Table 2 presents the measured and expected mass and apparent density of each lattice design. The expected mass was determined multiplying the volume fraction of the lattices by the density of PLA ($\rho_s = 1.24 \text{ g/cm}^3$), and the expected apparent density was calculated by dividing the expected mass by the cube volume. In addition, the percent deviations between the measured mass and expected mass values are displayed. A consistent trend towards smaller mass values and apparent densities for all lattice designs can be observed.

The deviation between the deposited and expected mass increased with the cell size in lattices designs with 0.4 mm and 0.6 mm wall thickness, whereas the opposite trend was observed for designs with 0.8 mm thickness. As depicted in the images of the printed structures, accurately reproducing thinner walls at the selected printing parameters proved more challenging, leading to the formation of holes, which were more likely to appear with larger cell units. Despite this general trend,

designs with 0.6 mm wall thickness exhibited the smallest deviations. In the case of designs with 0.8 mm thickness, which resulted in 3D printed structures free from wall holes, larger deviations with smaller cells may be related to the printing process's inability to precisely reproduce the curvature of the Gyroid lattices.

The better performance of the designs with 0.6 mm wall thickness was further investigated by analysing the deposition toolpaths generated by the slicer software, depicted in Fig. 5. The printing strategy involved depositing inner walls, followed by outer walls, and then filling the delimited area. In Gyroid lattices with 0.6 mm wall thickness, the deposition seems less continuous than in other topologies. The interruption of the deposition paths, linked to momentary printhead stops, may result in slight over-deposition. While this affects geometric accuracy negatively, it might have compensated for the general trend of under-depositing material per print.

The overall compression behaviour of the Gyroid lattices is represented in Fig. 6 for selected designs. Lattices featuring 10 mm cells exhibited successive layer collapse, as observed with the a10w06 design, indicating brittle behaviour. In contrast, lattices with 5 mm and 4 mm cells exhibited more plastic behaviour, with significant barrelling at the

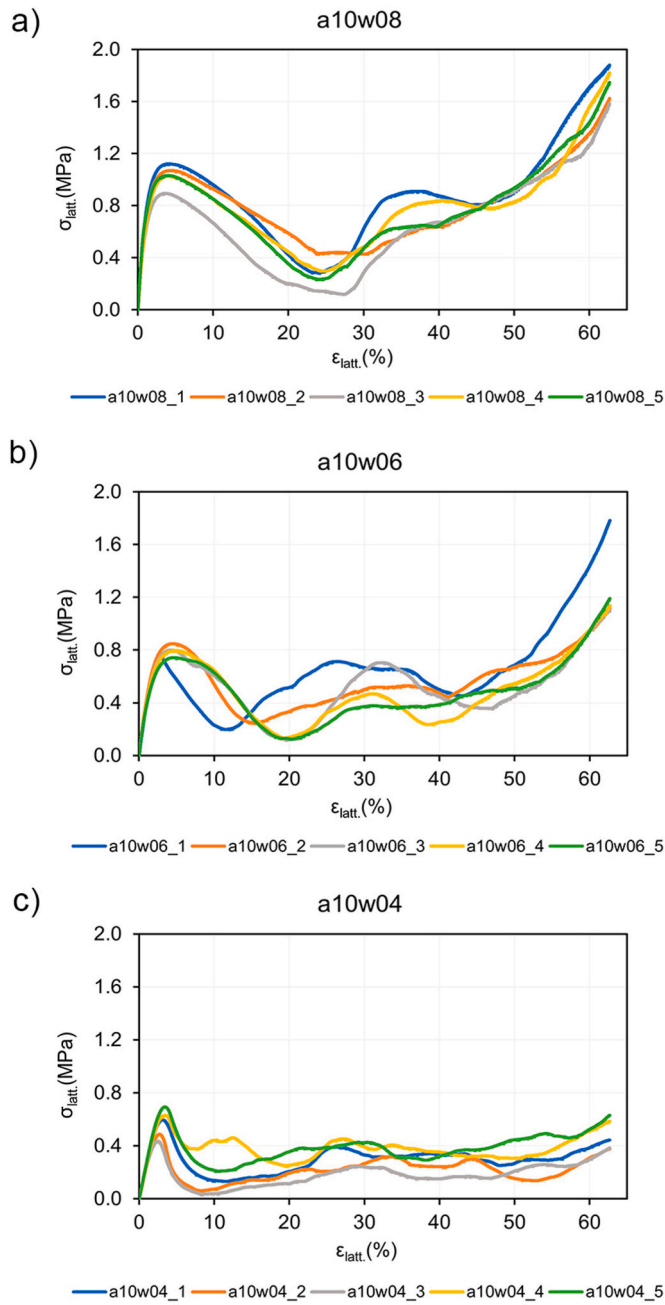


Fig. 7. Stress strain curves for the 10 mm cell Gyroid lattices with a) 0.8 mm wall thickness, b) 0.6 mm wall thickness, and c) 0.4 mm wall thickness.

intermediary layers as depicted in the a5w06 and a4w06 designs. These distinct behaviours, dependent on the cell size, are representative of all Gyroid lattices, except for the a5w04 design shown in Fig. 6, which also exhibited brittle fracture in several layers. The brittle fracture is evident from the twisting of the black painted lines marking the cells' edges within the structure. Such transition is clearly captured in the animations showing the deformation behaviour for all design combinations, available in the supplementary materials section.

The stress-strain curves resulting from the compression tests of the PLA Gyroid structures are shown through Figs. 7–9, according to the cell size. It is important to note that the graphs are not presented with the same scale in the stress axis across different cell sizes, because the span throughout the tested lattice designs was too wide. Nevertheless, the brittle and plastic behaviours discussed are clearly observed. The structures with 10 mm cells failed by sudden stress drops followed by

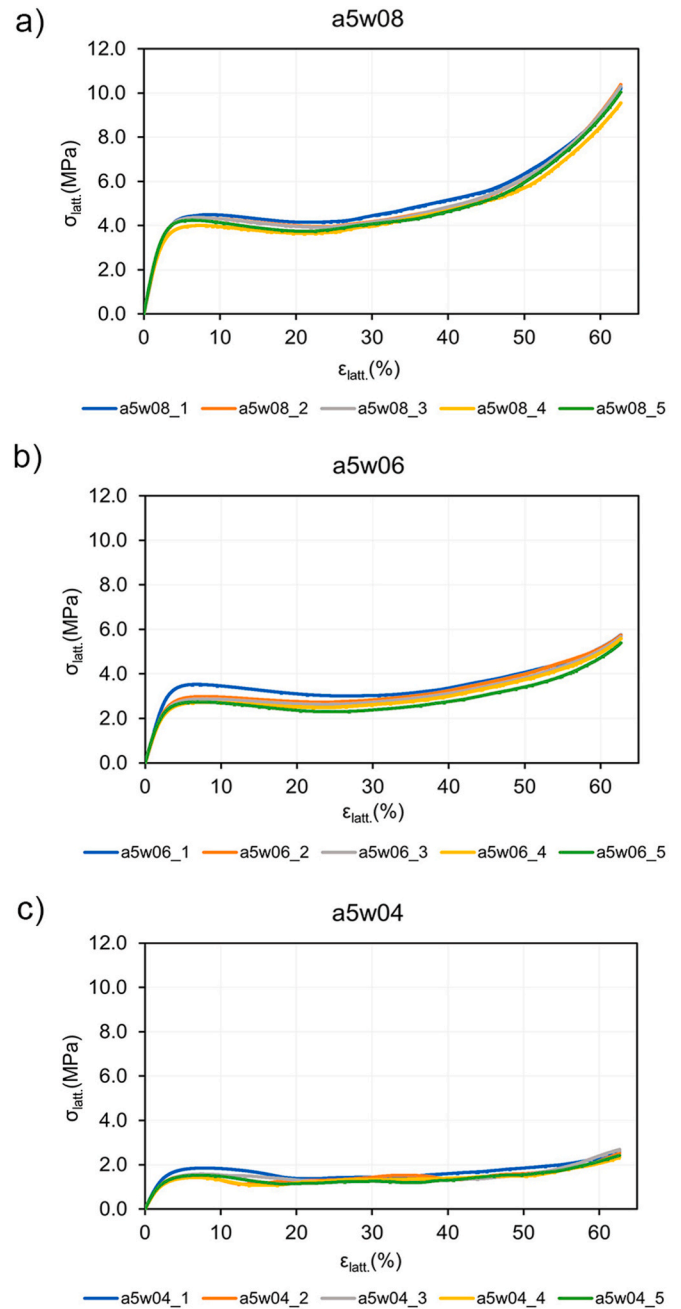


Fig. 8. Stress strain curves for the 5 mm cell Gyroid lattices with a) 0.8 mm wall thickness, b) 0.6 mm wall thickness, and c) 0.4 mm wall thickness.

densification regions, corresponding to the successive collapse of the lattice layers. The structures with 5 mm and 4 mm cells behaved as typical cellular materials, with an initial linear elastic region, a plateau and a final densification region. As mentioned, the only exception was the a5w04 design, which showed brittle behaviour to some extent. This can be observed by the presence of small peaks in the stress-strain curves, indicative of brittle layer collapse and densification. Overall, the variability of the stress-strain curves increased with decreasing values of the wall thickness, especially for the structures with 10 mm cells. As pointed in the analysis of the 3D printed samples, this variability may be attributed to the FFF process's inability to accurately reproduce the thinner Gyroid surfaces, resulting in the formation of wall holes where cracks may start.

Table 3 summarizes the elastic modulus, yield stress and strain, and absorbed energy for each lattice design alongside its corresponding

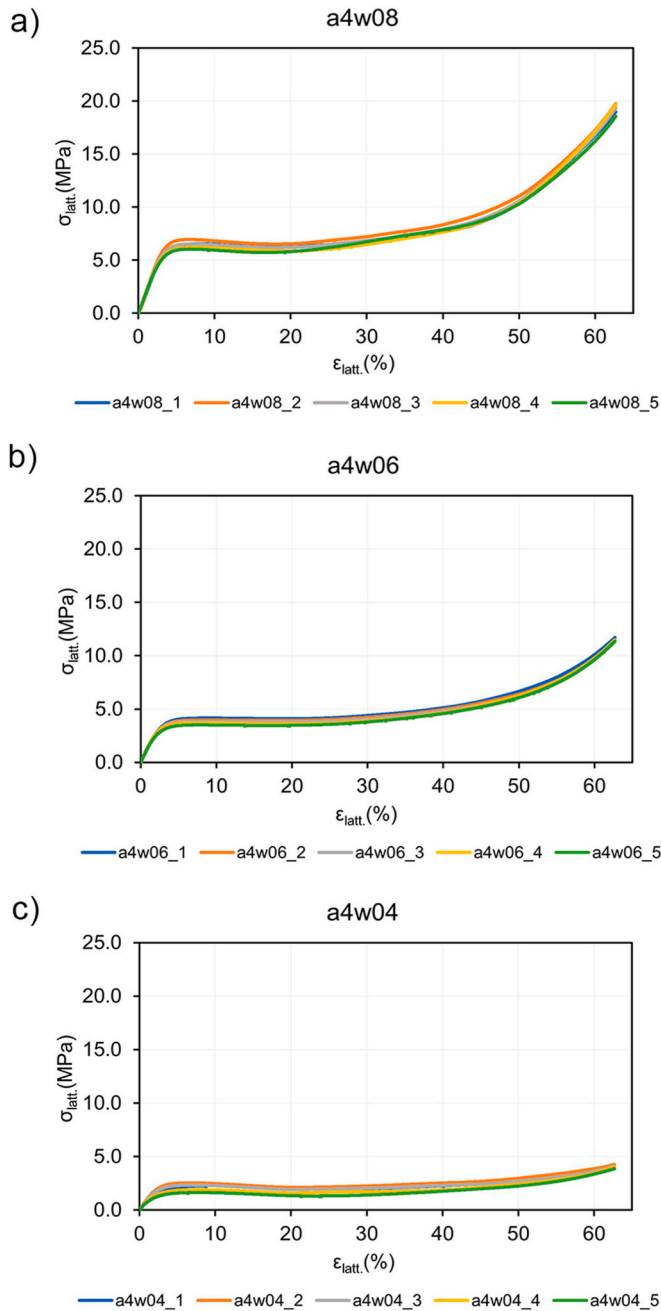


Fig. 9. Stress strain curves for the 4 mm cell Gyroid lattices with a) 0.8 mm wall thickness, b) 0.6 mm wall thickness, and c) 0.4 mm wall thickness.

apparent and relative density. The elastic modulus (E) was computed at 0.2% strain, and the yield point (σ_y , ϵ_y) was identified at the initial inflexion of the stress-strain curve. The absorbed energy was determined by calculating the area under each stress-strain curve. The a5w08, a4w08 and a4w06 designs exhibited the highest elastic modulus, measuring 165.54 ± 8.88 MPa, 162.05 ± 13.60 MPa and 143.79 ± 2.45 MPa, respectively. For the yield stress, the a4w08 lattice achieved the highest value with 6.43 ± 0.34 MPa. The highest amount of energy was absorbed by the a4w08 design, achieving 5.34 ± 0.15 MJ/m³. The a10w04 lattices showed the smallest elastic modulus, yield stress and absorbed energy, registering 25.96 ± 1.13 MPa, 0.44 ± 0.1 MPa and 0.18 ± 0.05 MJ/m³, respectively. These results can be correlated with the apparent density and volume fraction of the structures shown in the table, where denser designs generally exhibit better properties in agreement with (Maharjan et al., 2018). The Gyroid lattices with 4 mm

Table 3

Apparent density, volume fraction, elastic modulus, yield stress, yield strain, and energy absorbed for the PLA Gyroid lattices.

Lattice Design	ρ^* (g/cm ³)	Volume fraction	E (MPa)	σ_y (MPa)	ϵ_y (%)	Absorbed Energy (MJ/m ³)
a10w08	0.21	0.17	82.08 ± 7.52	1.03 ± 0.08	4.3 ± 0.4	0.49 ± 0.05
a10w06	0.16	0.13	46.50 ± 3.34	0.79 ± 0.04	4.3 ± 0.8	0.34 ± 0.04
a10w04	0.10	0.09	25.96 ± 1.13	0.44 ± 0.1	2.6 ± 0.4	0.18 ± 0.05
a5w08	0.38	0.33	165.54 ± 8.88	4.29 ± 0.17	7.6 ± 0.7	3.13 ± 0.10
a5w06	0.31	0.26	110.11 ± 8.32	2.96 ± 0.30	7.9 ± 0.6	2.00 ± 0.13
a5w04	0.20	0.17	65.01 ± 3.67	1.61 ± 0.15	7.2 ± 0.6	0.93 ± 0.06
a4w08	0.46	0.40	162.05 ± 13.60	6.43 ± 0.34	7.3 ± 0.4	5.24 ± 0.15
a4w06	0.39	0.32	143.79 ± 2.45	3.84 ± 0.22	8.5 ± 0.5	3.14 ± 0.12
a4w04	0.26	0.22	91.07 ± 10.55	2.13 ± 0.33	8.4 ± 1.4	1.39 ± 0.17

and 5 mm cells exhibited the highest yield strain, regardless of the wall thickness, when considering the standard deviations. In contrast, the lattices with 10 mm cells showed the smallest yield strains, with the a10w04 lattice exhibiting the smallest value at $2.6 \pm 0.4\%$ contradicting the numerical results found by (Yang et al., 2019).

As expected, the main effects plots in Fig. 10 indicate that when keeping the wall thickness constant, larger unit cell sizes resulted in decreased elastic modulus, yield stress, yield strain, and absorbed energy. The reduction in yield strain was particularly significant when transitioning from 5 mm to 10 mm unit cells. In terms of the wall thickness impact on lattices of identical unit cell sizes, the elastic modulus, yield stress, and absorbed energy exhibited an increase with thicker walls. The yielding point occurred at higher strains when transitioning from a 0.4 mm–0.6 mm wall thickness, but decreased when moving from 0.6 mm to 0.8 mm wall thickness.

In the interaction plots depicted in Fig. 11, the proportional decrease in elastic modulus with larger unit cell sizes appeared relatively consistent across varying wall thicknesses. The reduction in yield stress and absorbed energy was more pronounced in the lattices featuring a 0.8 mm wall thickness. Moreover, the decline in yield stress and absorbed energy was notably steeper when transitioning from 5 mm to 10 mm cells. With respect to the yield strain, lattices with 4 mm and 5 mm cells displayed similar means regardless of wall thickness, whereas a substantial decrease was observable only when scaling up from 5 mm to 10 mm cells.

Overall, the analysis suggests that variations in cell size exert a more significant influence on the mechanical results than changes in wall thickness. This phenomenon may be attributed to the observed transition in failure mode, shifting from predominantly plastic to predominantly brittle behaviour with larger cells. These two failure modes have also been observed in prior work by Maharjan et al. (2018) involving PLA Gyroids. However, in that case, the influence of volume fraction, and consequently wall thickness, was predominant. In the results reported by Maconachie et al. (2020) regarding ABS Gyroids, even if both brittle layer fracture and plastic barrelling were observed for low-density and high-density lattices, respectively, halving the number

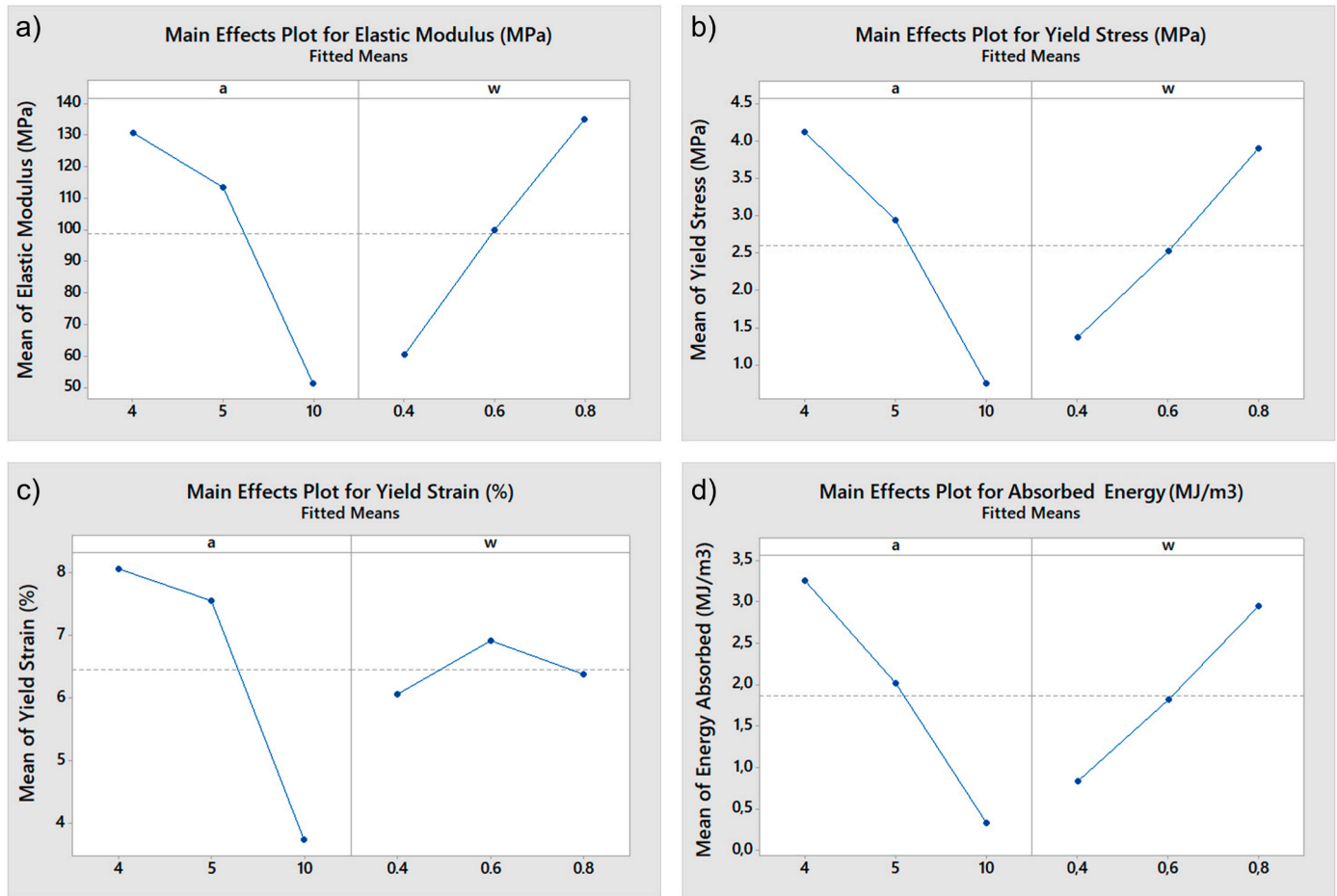


Fig. 10. Main effects plots of unit cell size (a) and wall thickness (w) in the: a) Elastic Modulus, b) Yield Stress, c) Yield Strain, and d) Absorbed Energy.

of cells or doubling the wall thickness appears to yield approximately the same effect on the mechanical properties. Although it was not discussed by Holmes et al. (2022), the influence of wall thickness seemed to be more important than the cell size for the compressive properties of thermoplastic polyurethane.

The specific properties of the Gyroid lattices, calculated with reference to their apparent density, are exhibited in Fig. 12. In terms of specific elastic modulus under compression, most designs stood around $350 \times 10^6 \text{ (m}^2/\text{s}^2)$. The highest values were given by the a5w08 lattices, with $435.6 \pm 23.4 \times 10^6 \text{ (m}^2/\text{s}^2)$, and the a10w08 lattices, with $390.9 \pm 35.8 \times 10^6 \text{ (m}^2/\text{s}^2)$, despite not being the densest structures. Conversely, the lowest specific modulus was observed with the a10w04 design, with $259.6 \pm 11.3 \times 10^6 \text{ (m}^2/\text{s}^2)$, aligning with their lower apparent density. The highest values for the specific yield stress were observed with the a4w08 design, achieving $14 \pm 0.7 \times 10^3 \text{ (m}^2/\text{s}^2)$, while the lowest were given by a10w04 lattices, with $4.4 \pm 1 \times 10^3 \text{ (m}^2/\text{s}^2)$. The best performance in terms of specific energy absorbed was also achieved by the a4w08 lattices, with $11.4 \times 10^3 \text{ (m}^2/\text{s}^2)$, and the worst was achieved by the a10w04 lattices, with $1.8 \times 10^3 \text{ (m}^2/\text{s}^2)$. Given the potential lightweight applications for such materials, structures featuring 4 mm cells and 0.8 mm walls are suggested to deliver better performance across the tested properties.

The relative modulus (E^*/E_s) and relative yield stress (σ_y^*/σ_{ys}) of the PLA Gyroid lattices, calculated in function of their relative density (ρ^*/ρ_s), are presented in Fig. 13. The values for E_s and σ_{ys} ($1309.97 \pm 52.59 \text{ MPa}$ and $52.82 \pm 6.25 \text{ MPa}$, respectively) were determined by testing solid PLA 20 mm cubes, which were 3D printed and compressed under the same conditions as the lattice structures. The graphs were generated using the Granta EduPack® 2023 R1 software. The dotted

lines representing the ideal behaviour of stretch- and bending-dominated lattices were drawn with the slopes provided by Ashby (2006).

The graphs also shows that the lattice designs with higher relative density exhibited better performance in terms of relative stiffness and yield stress. However, among design combinations with similar relative densities (represented by a10w08 and a5w04), larger cells and thicker walls appear to lead to a higher relative elastic modulus, while smaller cells and thinner walls seem to result in higher relative yield stress. Considering the fitted trend lines, the slope calculated for the relative yield stress is closer to that expected for ideal bending-dominated structures (1.78 compared to 1.5, respectively) than the slope calculated for the relative elastic modulus (1.24 compared to 2). Regardless, the overall behaviour of relative stiffness and yield stress of the Gyroid lattices tested can be considered similar to that of polyurethane foams with closed cells.

4. Conclusions

In this study, a parametric design algorithm was developed to promptly create Gyroid lattices with different cell sizes (a) and wall thicknesses (w). The cell size and wall thickness were varied in three levels: 10 mm, 5 mm, and 4 mm, and 0.8 mm, 0.6 mm, and 0.4 mm, respectively. The designs were 3D printed using FFF with PLA and subjected to compression testing.

Despite the suitability of the FFF process for fabricating intricate geometries, a general trend towards smaller mass and apparent density values was observed across all lattice designs. The deviations in mass were more significant in designs with larger cells and thinner walls,

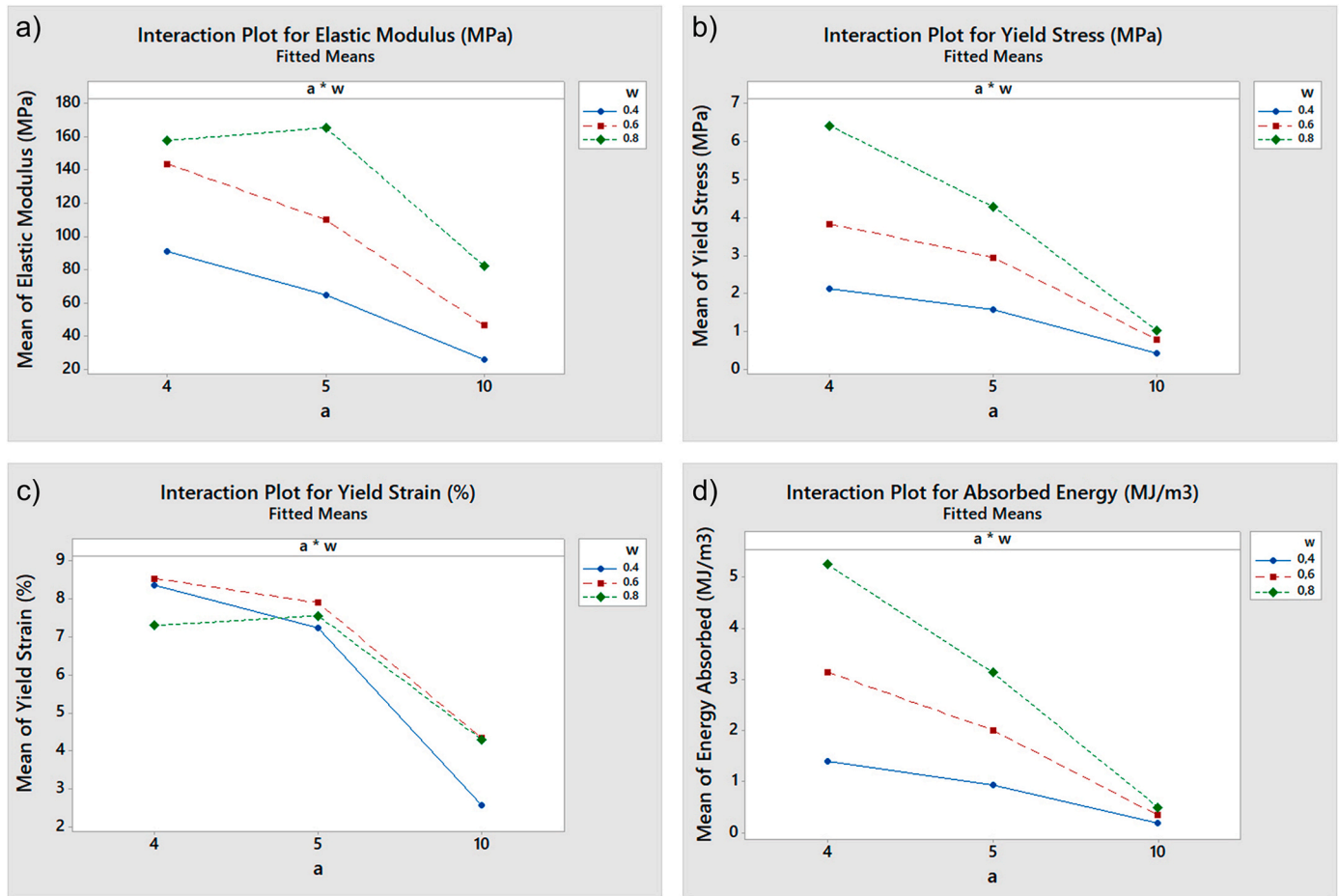


Fig. 11. Interaction plots between unit cell size (a) and wall thickness (w) in the: a) Elastic Modulus, b) Yield Stress, c) Yield Strain, and d) Energy Absorbed.

which might be related to the wall holes visualized in the printed structures. These holes were formed mostly in unsupported surfaces, due to the inability of the FFF process to reproduce these geometries at the prescribed settings. The difference between the expected and measured mass were less severe in lattices with a 0.6 mm wall thickness, which could have been compensated by residual material deposition due to the many interruptions in the toolpaths generated by the slicer software.

During compression testing, the lattice designs with 10 mm unit cells exhibited successive collapse of Gyroid layers in a more brittle behaviour. Structures with 4 mm and 5 mm cells deformed more plastically, exhibiting the typical behaviour of bending-dominated cellular materials. These two behaviours were independent from the wall thickness, except for the a5w04 design, which also showed brittle behaviour to some extent. Both failure behaviours could be observed in the stress-strain plots of the tested structures. Brittle layer collapse and densification was associated with multiple peaks in the resulting curves, while plastic deformation was marked by an initial plateau followed by final densification.

The a5w08, a4w08, and a4w06 designs exhibited the highest elastic moduli, with values around 165 MPa. The a4w08 design also presented the highest yield stress (approximately 6.5 MPa) and energy absorbed per volume (around 5.3 MJ/m³). In turn, the a10w04 design presented the lowest mechanical properties, registering approximately 26 MPa for the elastic modulus, 0.4 MPa for the yield stress, and 0.2 MJ/m³ for the absorbed energy. Overall, in terms of elastic modulus and yield stress, the resulting properties range between 1% and 12% of the bulk 3D printed material. For the energy absorbed per volume, the results are between 1% and 22%.

As a rule, any decrease in the apparent density caused by larger unit

cells or thinner walls resulted in decreased stiffness, yield strength and energy absorption. For structures with similar apparent density, larger cells and thicker walls resulted in higher stiffness and lower yield strength and absorbed energy. Considering the isolated factors, doubling the cell size affected more the measured properties than doubling the wall thickness. With respect to the interaction between the tested factors, doubling the cell size and wall thickness had more influence on the yield stress and yield strain.

While categorizing the Gyroid lattices into open or closed cell foams poses challenges due to their 3D curved surfaces that form interconnected channels, the observed relationship between the relative density and the w/a ratio approximates to that of closed-cells foams. Similarly, the mechanical properties under compression exhibited behaviour akin to foams with closed cells under the conditions tested in this work. Future studies can explore whether similar behaviour can be observed independent of the properties of the parent materials and 3D printing process.

CRediT authorship contribution statement

Joaquim Justino Netto: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Manuel Sardinha:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Marco Leite:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

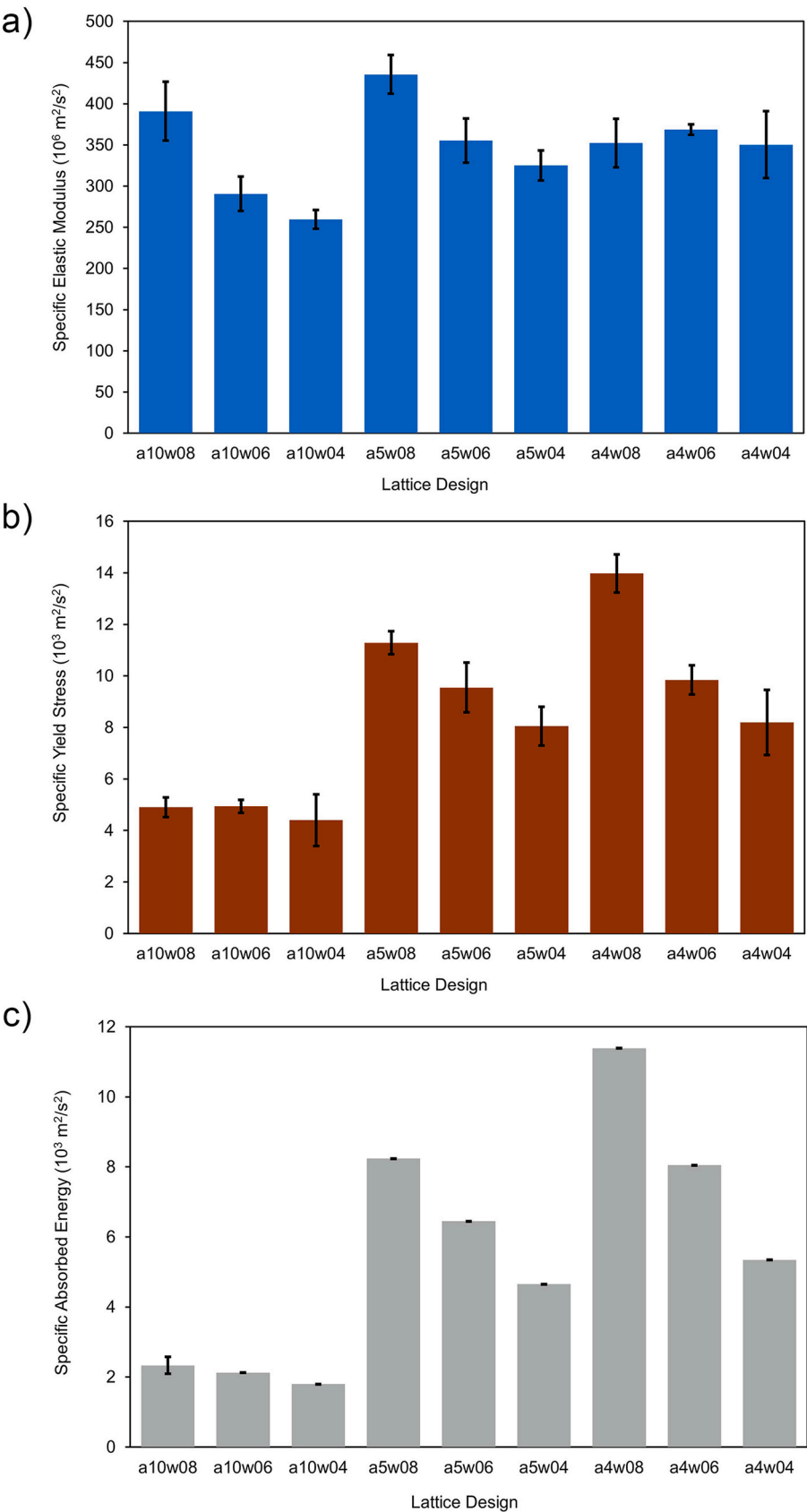


Fig. 12. Mechanical properties of the Gyroid lattices with respect to their apparent density: (a) specific elastic modulus, (b) specific yield stress, and (c) specific energy absorbed.

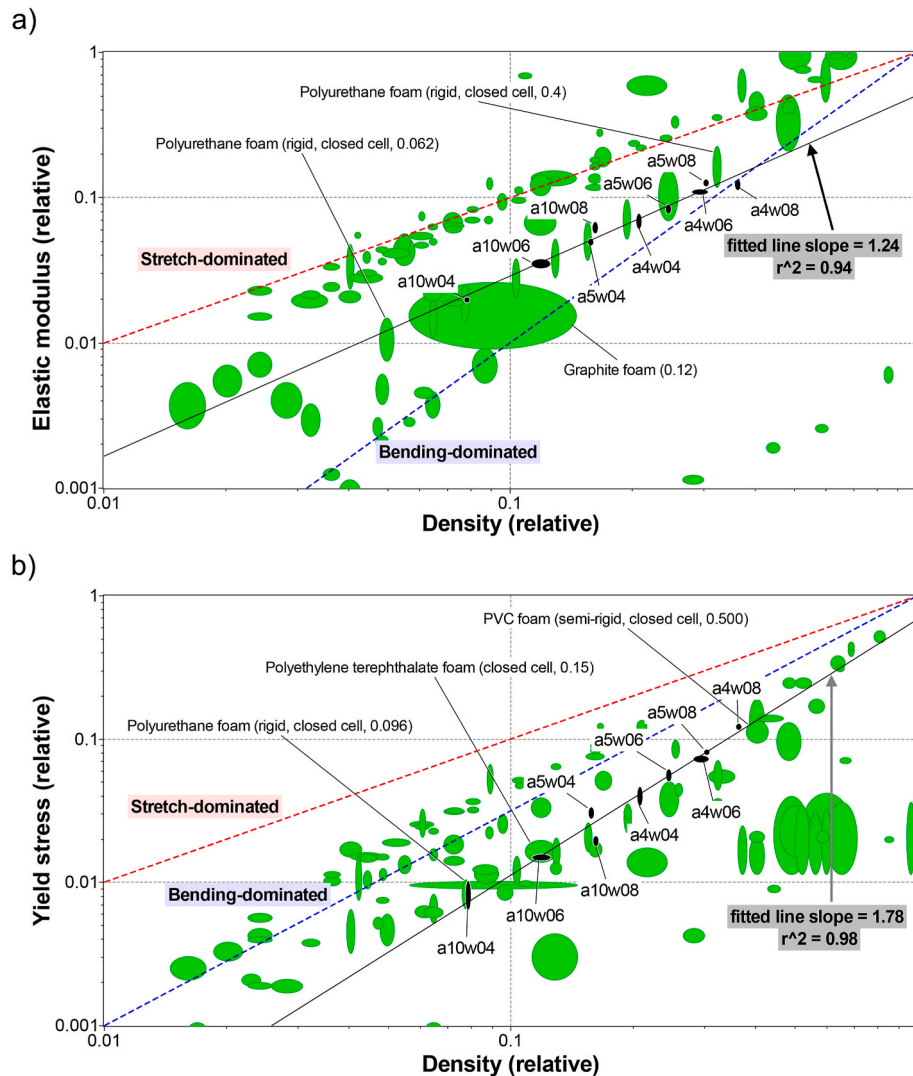


Fig. 13. Relative mechanical properties of the PLA Gyroid lattice: a) Relative elastic modulus, and b) Relative yield stress in function of the relative density.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT 3.5 in order to improve the language level. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mechmat.2024.105051>.

References

- Agnelli, J., Pagano, C., Fassi, I., Treccani, L., Bignotti, F., Baldi, F., 2023. Mechanical behaviour of ductile polymer cellular model structures manufactured by FDM. *Mech. Mater.*, 104882 <https://doi.org/10.1016/j.mechmat.2023.104882>.
- Ashby, M.F., 2006. The properties of foams and lattices. *Phil. Trans. Math. Phys. Eng. Sci.* 364 (1838), 15–30. <https://doi.org/10.1098/rsta.2005.1678>.
- Ashby, M.F., 2011. Designing hybrid materials. In: *Materials Selection in Mechanical Design*. Elsevier, pp. 299–340. <https://doi.org/10.1016/b978-1-85617-663-7.00011-4>.

- Bayati, A., et al., 2024. 3D printing super stretchable propylene-based elastomer. *Mater. Lett.* 361 (Apr) <https://doi.org/10.1016/j.matlet.2024.136075>.
- Benedetti, M., du Plessis, A., Ritchie, R.O., Dallago, M., Razavi, N., Berto, F., 2021. Architected cellular materials: a review on their mechanical properties towards fatigue-tolerant design and fabrication. *Mater. Sci. Eng. R Rep.* 144 <https://doi.org/10.1016/j.mser.2021.100606>. Elsevier Ltd.
- Bonner-Hutton, O., et al., 2023. Analysis of gyroid heat exchangers for superconducting electric motors. *Mater. Today Proc.* <https://doi.org/10.1016/j.matpr.2023.03.780>.
- Bouzaglou, O., Golan, O., Lachman, N., 2023. Process design and parameters interaction in material extrusion 3D printing: a review. *Polymers* 15 (10). <https://doi.org/10.3390/polym15102280>. MDPI.
- Chouhan, G., Bala Murali, G., 2023. Designs, advancements, and applications of three-dimensional printed gyroid structures: a review. *Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering*. SAGE Publications Ltd. <https://doi.org/10.1177/09544089231160030>.
- Costa Oliveira, F.A., et al., 2023. Manufacturing and thermal shock resistance of 3D-printed porous black zirconia for concentrated solar applications. *Crystals* 13 (9). <https://doi.org/10.3390/cryst13091323>.
- Evans, A.G., Hutchinson, J.W., Fleck, N.A., Ashby, M.F., Wadley, H.N.G., 2001. The topological design of multifunctional cellular metals. *Prog. Mater. Sci.* 46 (3–4), 309–327. [https://doi.org/10.1016/S0079-6425\(00\)00016-5](https://doi.org/10.1016/S0079-6425(00)00016-5).
- Holmes, D.W., et al., 2022. Mechanical behaviour of flexible 3D printed gyroid structures as a tuneable replacement for soft padding foam. *Addit. Manuf.* 50 (Feb) <https://doi.org/10.1016/j.addma.2021.102555>.
- ISO/ASTM 52900, 2015. Standard terminology for additive manufacturing – general principles – terminology [Online]. Available: <http://www.ansi.org>.
- Kelly, C.N., Kahra, C., Maier, H.J., Gall, K., 2021. Processing, structure, and properties of additively manufactured titanium scaffolds with gyroid-sheet architecture. *Addit. Manuf.* 41 <https://doi.org/10.1016/j.addma.2021.101916>.
- Maconachie, T., et al., 2020. The compressive behaviour of ABS gyroid lattice structures manufactured by fused deposition modelling. *Int. J. Adv. Manuf. Technol.* 107 (11–12), 4449–4467. <https://doi.org/10.1007/s00170-020-05239-4>.
- Maharjan, G.K., Khan, S.Z., Riza, S.H., Masood, S.H., 2018. Compressive behaviour of 3D printed polymeric gyroid cellular lattice structure. In: *IOP Conference Series: Materials Science and Engineering*. Institute of Physics Publishing. <https://doi.org/10.1088/1757-899X/455/1/012047>.
- Mahmoud, D., Al-Rubaie, K.S., Elbestawi, M.A., 2021. The influence of selective laser melting defects on the fatigue properties of Ti6Al4V porosity graded gyroids for bone implants. *Int. J. Mech. Sci.* 193 <https://doi.org/10.1016/j.ijmecsci.2020.106180>.
- Makatura, L., et al., 2023. Procedural metamaterials: a unified procedural graph for metamaterial design. *ACM Trans. Graph.* 42 (5), 1–19. <https://doi.org/10.1145/3605389>.
- Maskery, I., Aboulkhair, N.T., Aremu, A.O., Tuck, C.J., Ashcroft, I.A., 2017. Compressive failure modes and energy absorption in additively manufactured double gyroid lattices. *Addit. Manuf.* 16, 24–29. <https://doi.org/10.1016/j.addma.2017.04.003>.
- Nazir, A., Abate, K.M., Kumar, A., Jeng, J.Y., 2019. A state-of-the-art review on types, design, optimization, and additive manufacturing of cellular structures. *Int. J. Adv. Manuf. Technol.* 104 (9–12), 3489–3510. <https://doi.org/10.1007/s00170-019-04085-3>.
- Oliveira, A., et al., 2022. Evaluation of cellular structures with triply periodic minimal surfaces fabricated by additive manufacturing. *Eng. Manufacturing Letters* 1 (1), 28–33. https://doi.org/10.24840/2795-5168_001-001_0006.
- Tino, R., Leary, M., Yeo, A., Brandt, M., Kron, T., 2019. Gyroid structures for 3D-printed heterogeneous radiotherapy phantoms. *Phys. Med. Biol.* 64 (21) <https://doi.org/10.1088/1361-6560/ab48ab>.
- Wallat, L., Altschuh, P., Reder, M., Nestler, B., Poehler, F., 2022. Computational design and characterisation of gyroid structures with different gradient functions for porosity adjustment. *Materials* 15 (10). <https://doi.org/10.3390/ma15103730>.
- Yang, E., et al., 2019. Effect of geometry on the mechanical properties of Ti-6Al-4V Gyroid structures fabricated via SLM: a numerical study. *Mater. Des.* 184 (Dec) <https://doi.org/10.1016/j.matdes.2019.108165>.
- Zadpoor, A.A., 2019. Mechanical performance of additively manufactured meta-biomaterials. *Acta Biomater.* 85 <https://doi.org/10.1016/j.actbio.2018.12.038>. Acta Materialia Inc, pp. 41–59, Feb. 01.
- Zhu, X., Chen, F., He, N., Han, X., 2022. The design and fabrication of gyroid scaffolds for breast reconstruction by fused deposition modelling. *Mater. Today Proc.* 70, 119–123. <https://doi.org/10.1016/j.matpr.2022.08.558>.